

Light Energy — Workout

ICSE Class 7 Physics · 10 MCQs · 1 mark each · Name: _____ Date: _____

Syllabus: Whole chapter: reflection and its laws, plane-mirror images, regular/diffuse reflection, speed of light, and the addition & subtraction of colours.

Instructions: Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.

1. A ray strikes a plane mirror perpendicularly. It retraces its path because the angle of incidence is:

- | | |
|---------------|----------------|
| A. 90° | B. 180° |
| C. 0° | D. 45° |

2. An object is 25 cm from a plane mirror. The image is ____ from the object.

- | | |
|------------|----------|
| A. 50 cm | B. 75 cm |
| C. 12.5 cm | D. 25 cm |

3. Two plane mirrors face each other parallel (angle 0°). The number of images formed is:

- | | |
|-------------|--------|
| A. infinite | B. two |
| C. none | D. one |

4. In a mirror a wall clock appears to show 9:00. The real time is:

- | | |
|----------|---------|
| A. 12:00 | B. 9:00 |
| C. 6:00 | D. 3:00 |

5. Light falling on a white wall is reflected diffusely, which is why the wall:

- | | |
|---------------------|--|
| A. acts as a mirror | B. is visible from all directions but shows no image |
| C. cannot be seen | D. appears black |

6. Which of these is a virtual image?

- | | |
|---------------------------------------|-----------------------------------|
| A. the image on a wall from a pinhole | B. the image in a bathroom mirror |
| C. the image on a cinema screen | D. the image on photographic film |

7. Light from the Moon takes about 1.3 s to reach Earth. Taking $c = 3 \times 10^8$ m/s, the distance is about:

- | | |
|------------------------|------------------------|
| A. 3.9×10^8 m | B. 2.3×10^8 m |
| C. 1.3×10^8 m | D. 4×10^5 m |

8. On a white screen, overlapping blue and green spotlights give:

- | | |
|------------|----------|
| A. magenta | B. cyan |
| C. yellow | D. white |

9. A white shirt seen under red stage lighting appears:

- | | |
|----------|----------|
| A. black | B. white |
| C. green | D. red |

10. A yellow object is viewed through a blue filter (only blue passes). The object appears:

- | | |
|-----------|----------|
| A. yellow | B. green |
| C. black | D. blue |

Answer Key

1. C

2. A

3. A

4. D

5. B

6. B

7. A

8. B

9. D

10. C